MODULE 13 · DOMAIN 3 - INTEGRATION

Choosing the Integration Mechanism

A direct API or CLI call, an MCP server, or a hand-off to another agent: how to read the brief, and what each choice commits you to.

DOMAIN 3 of 7 EXAM WEIGHT 19% MODULE 13 of 38

admin@sthithapragna.com

A direct phone line, a universal power socket, or a colleague who **thinks for themselves**.

admin@sthithapragna.com

Three mechanisms, three profiles

MECHANISM	CHOOSE IT WHEN	WHAT IT COSTS
Direct API or CLI	One consumer, a stable endpoint, stateless, latency-sensitive	Each new client rebuilds the integration
MCP server	Several clients or hosts need the same capability	A server to build, secure and run
Agent-to-agent	Another autonomous agent owns the work: long-running, stateful	Another model in the path: latency, cost, tracing

ON THE EXAM

Start from the simplest mechanism that meets the brief. Each step up must be bought by a stated need.

Reading the brief

SIGNAL IN THE BRIEF	LEANS TOWARDS
Called on every reply, answers in milliseconds	Direct API
A trusted CLI already exists for the job	Direct CLI
Claude Code, an app and an agent all need it	MCP
The vendor already publishes an MCP server	MCP
The other side plans, decides and runs for hours	Agent-to-agent
It is a procedure, not access to a system	A Skill, not an integration

THE TIEBREAKER

Count the consumers, time the call, and ask who does the thinking.

MCP in one slide

- **Host, client, server:** the app runs a client per server; the server exposes capabilities.
- **Tools** are actions the model chooses to call.
- **Resources** are data the application attaches as context.
- **Prompts** are templates a user picks, such as a slash command.
- **Write once:** Claude Code, Claude apps, the Agent SDK and the API connector can all use it.

> ON THE EXAM
>
> MCP standardises the socket. It does not decide who may plug in. That is still your job.

Local or remote: stdio or Streamable HTTP

STDIO

A local process

- Runs on the user's own machine
- Credentials come from the environment
- One user, no network hop
- Each machine holds its own copy
- Hard to revoke, update or audit centrally

STREAMABLE HTTP

A remote service

- Hosted once, used by many clients
- OAuth 2.1 authorisation, per user
- Central logging, updates and revocation
- Scopes set by the server
- Needs hosting and security work

THE TIEBREAKER

One developer's local tool: stdio. Shared, audited, used by many: remote over Streamable HTTP.

Authorisation on a remote MCP server

Authorisation server

Signs the user in
Issues the tokens

MCP client

PKCE and resource indicator
Requests minimal scopes

MCP server

Checks each token was issued for it
Accepts no other tokens

An OAuth 2.1 resource server

Downstream APIs

Separate token, own audience
Never the client's token

Discovery

Protected Resource Metadata
Scopes in the challenge

THE TAKEAWAY

No token passthrough: a server must not accept or pass on tokens that were issued for anyone else.

Why token passthrough is forbidden

- **Controls are skipped:** rate limits and validation in the server never see the real request.
- **The audit trail breaks:** downstream logs cannot tell who really acted, or through what.
- **Trust boundaries blur:** a token meant for one service starts working at another.
- **The fix:** validate the audience inbound; get a separate token for each downstream call.

> THE PICTURE
>
> A server that forwards whatever it is handed becomes a confused deputy.

MCP today: the 2026-07-28 specification

CHANGE	WHAT IT MEANS
Stateless protocol	No initialise handshake, no sessions
server/discover	Advertises versions and capabilities
Cross-call state	Explicit handles, passed as tool arguments
Deprecated	Sampling, roots, logging, HTTP+SSE
Client registration	Metadata documents replace dynamic registration
Tasks	Moved out of the core, into an extension

ON THE EXAM

The architecture is unchanged: tools, resources and prompts, over stdio or Streamable HTTP, with OAuth on HTTP.

The MCP connector on the Messages API

QUESTION	ANSWER, OCTOBER 2026
What it does	Calls remote MCP servers with no client of your own
Which servers	Public HTTPS only. No stdio
Which features	Tool calls only
Authentication	You run OAuth and pass the token
Tool control	Allow-list or deny-list per server
Where and status	Beta. Claude API, Claude Platform on AWS, Foundry. Not ZDR

ON THE EXAM

Not listed for Bedrock or Google Cloud. There, run the MCP client in your own application.

Agent-to-agent

- **Use it when** another autonomous agent owns the outcome, not just a lookup.
- **Typical shape:** long-running, stateful work with status, questions back, and a result.
- **Often across boundaries:** different teams, vendors or organisations.
- **A2A** is one industry protocol: agent cards for discovery, tasks with a lifecycle.
- **Never for a lookup:** a model hop costs latency, tokens and traceability.

> ON THE EXAM
>
> The exam tests the pattern, not any one protocol. Ask who owns the thinking on the other side.

> Tallyhawk's help-centre search is needed by the support agent, by engineers in Claude Code, and by the customer success team in Claude apps.
>
> Each team built its own integration. The three versions now return different results for the same query.

What should Corvane recommend?

A Keep three integrations, built on a shared code library

B Load the help articles into each system prompt instead

C Build one remote MCP server for help-centre search, used by all three

D Create a help-centre agent that the others hand questions to

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Keep three integrations, built on a shared code library

Still three hosts to wire up, and three places to drift.

B Load the help articles into each system prompt instead

Nine thousand articles: far too large, and stale on every edit.

C Build one remote MCP server for help-centre search, used by all three

Three hosts, one capability: the case MCP exists for.

D Create a help-centre agent that the others hand questions to

A model hop for a search. The other side does no thinking.

THE SIGNAL

Several hosts need one capability, and copies drift: one MCP server.

> Tallyhawk engineers want Claude Code to check deployment status and recent releases. Every developer machine already has tallyctl, the company's well-documented internal CLI.
>
> A platform engineer proposes building an MCP server that wraps the deployment API.

What should Corvane recommend first?

A Build the MCP server as proposed, for a consistent interface

B Paste recent deployment logs into CLAUDE.md each morning

C Write a Skill that explains how deployments work

D Let Claude Code run tallyctl, allowing its read-only status commands

◆ PAUSE AND DECIDE

What should Corvane recommend first?

A Build the MCP server as proposed, for a consistent interface

A new server to build and run, duplicating a tool that works.

B Paste recent deployment logs into CLAUDE.md each morning

Stale within hours, and tokens on every session.

C Write a Skill that explains how deployments work

Knowledge, not access. It cannot read live status.

D Let Claude Code run tallyctl, allowing its read-only status commands

An existing, trusted CLI: the most context-efficient path.

THE SIGNAL

A trusted CLI already does the job: let Claude use it, scoped to what is needed.

> Mereside's mortgage agent needs property valuations from Halden Survey Partners, an independent surveyor. Halden runs its own valuation agent.
>
> That agent books inspections, asks questions back, and returns a signed report, often days later. Halden will not give Mereside access to its internal systems.

Which integration should Corvane design?

A Hand each request to Halden's agent agent-to-agent, and track the task until the report returns

B Call Halden's booking API synchronously, and wait for the report

C Give Mereside's agent tools into Halden's internal booking systems

D Automate Halden's web portal with a browser every night

◆ PAUSE AND DECIDE

Which integration should Corvane design?

A Hand each request to Halden's agent agent-to-agent, and track the task until the report returns

Another organisation's agent owns long-running, stateful work.

B Call Halden's booking API synchronously, and wait for the report

A synchronous call cannot wait days, or answer questions back.

C Give Mereside's agent tools into Halden's internal booking systems

Halden has refused, and its process belongs to its own agent.

D Automate Halden's web portal with a browser every night

Brittle, unagreed, and still cannot answer questions back.

THE SIGNAL

Another organisation's agent, days of stateful work, questions back: agent-to-agent.

> Tallyhawk engineers each run a local stdio copy of an internal billing MCP server, all using one shared admin key from their environment.
>
> The server will now also serve the support agent and all 18 teams. Security wants central revocation and per-user audit.

What should Corvane recommend?

A. Keep the local copies, and rotate the shared key every month

B. Host it once over Streamable HTTP, with OAuth and per-user scopes

C. Give each client the billing key in its system prompt

D. Turn the server into a Skill with scripts that call billing

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Keep the local copies, and rotate the shared key every month

Still one shared admin key: no per-user audit, no clean revocation.

B. Host it once over Streamable HTTP, with OAuth and per-user scopes

Shared by many, audited per user, revoked in one place.

C. Give each client the billing key in its system prompt

A credential in model context. Never.

D. Turn the server into a Skill with scripts that call billing

Moves the shared key into scripts. Access still needs proper auth.

THE SIGNAL

Many users, central control, per-user audit: remote Streamable HTTP with OAuth.

> Quillon is building a remote MCP server over its health record. Clinicians' clients send it OAuth tokens issued for the MCP server.
>
> A developer proposes forwarding each incoming token to the health record API, so the record system sees which clinician is asking.

What should Corvane recommend?

A. Forward the incoming token as proposed, since it identifies the clinician

B. Call the health record API with one shared service account

C. Validate each token's audience, and get a separate token for the record API on the clinician's behalf

D. Pass the clinician's name in each tool argument instead

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Forward the incoming token as proposed, since it identifies the clinician

Token passthrough. Forbidden by the specification.

B Call the health record API with one shared service account

Loses the clinician's identity and limits. Audit cannot attribute access.

C Validate each token's audience, and get a separate token for the record API on the clinician's behalf

Each hop has its own token, and the clinician stays the actor.

D Pass the clinician's name in each tool argument instead

A claim anyone can type. Not authorisation.

THE SIGNAL

Identity must survive each hop, but each token belongs to one audience. No passthrough.

> Mereside uses Claude through Amazon Bedrock in a European region. Its complaints case system has an internal MCP server, reachable only inside Mereside's network.
>
> A developer plans to use the Messages API's MCP connector to reach it.

What should Corvane recommend?

A Use the MCP connector through the Bedrock endpoint, as planned

B Move this workload to the Claude API so the connector is available

C Expose the complaints MCP server publicly so the connector can reach it

D Run the MCP client in Mereside's own application, passing its tools to Bedrock

◆ PAUSE AND DECIDE

What should Corvane recommend?

A. Use the MCP connector through the Bedrock endpoint, as planned

The connector is not listed for Bedrock, and needs a public server.

B. Move this workload to the Claude API so the connector is available

Breaks the EU processing constraint that chose Bedrock.

C. Expose the complaints MCP server publicly so the connector can reach it

Opens a complaints system to the internet to suit a feature.

D. Run the MCP client in Mereside's own application, passing its tools to Bedrock

Works on any platform, and the server stays private.

THE SIGNAL

Hard constraints first: platform and network boundary. Then run the client yourself.

> Tallyhawk's engineering teams want Claude Code to read and update issues in Lanternly, their issue tracker. Lanternly publishes an official remote MCP server.
>
> One team has started writing its own MCP wrapper over Lanternly's API. Another found a community server online.

What should Corvane recommend?

A. Finish the in-house wrapper, so Tallyhawk controls the code

B. Adopt Lanternly's official server after a security review, enabling only the tools teams need

C. Use the community server, since MCP servers are interchangeable

D. Have each team call Lanternly's API from its own scripts

◆ PAUSE AND DECIDE

What should Corvane recommend?

A Finish the in-house wrapper, so Tallyhawk controls the code

Rebuilds and maintains what the vendor already supports.

B Adopt Lanternly's official server after a security review, enabling only the tools teams need

Vendor-maintained, reviewed, and trimmed to the tasks.

C Use the community server, since MCP servers are interchangeable

Unknown code with your credentials. Its annotations are untrusted.

D Have each team call Lanternly's API from its own scripts

Eighteen integrations to drift, the problem MCP removes.

THE SIGNAL

Prefer the maintained official server, reviewed and trimmed, over building or borrowing.

> Tallyhawk is deciding how to expose its new route-optimisation capability to Claude.
>
> Corvane lists five facts that might appear in the final brief.

Which two facts would make MCP the better choice over a direct tool call? Choose two.

A Claude Code, the support app and a partner's agent will all use it

B It is called on every reply and must answer in tens of milliseconds

C Only one internal agent will ever call it

D The optimiser plans and negotiates routes for hours on its own

E Customers want to connect their own Claude clients to it

◆ PAUSE AND DECIDE

Which two facts would make MCP the better choice over a direct tool call? Choose two.

A Claude Code, the support app and a partner's agent will all use it

Several hosts, one capability.

B It is called on every reply and must answer in tens of milliseconds

Points to a direct call.

C Only one internal agent will ever call it

One consumer. direct is simpler.

D The optimiser plans and negotiates routes for hours on its own

Points to agent-to-agent, not MCP.

E Customers want to connect their own Claude clients to it

Clients you do not build: a standard interface.

THE SIGNAL

Many consumers, or consumers you do not control: MCP. One fast consumer: direct.

In five lines

- **Direct API or CLI** for one consumer, a stable endpoint, and tight latency.
- **MCP** when several hosts, or clients you don't build, need one capability.
- **Agent-to-agent** when another autonomous agent owns long-running work.
- **Remote MCP:** Streamable HTTP, OAuth 2.1, per-user scopes, no token passthrough.
- **Check the platform:** the API's MCP connector is remote, tools-only, and not listed on Bedrock.

COMING UP

Next: RAG pipelines. Chunking, indexing and retrieval matched to the data.